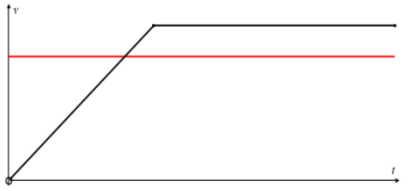
Question			Answer	Marks	AO	Guidance
8	(a)		6 5.4  5	B1 B1 B1	3.3 3.3 3.3	Straight line segment with positive gradient from the origin Subsequent line segment horizontal and above the given line Gradient change in their graph labelled with 5 and 6. (May be on axes)
				[3]		
8	(b)		velocity of the boy [$t < 5$ s] $v = 1.2t = 5.4$ giving $t = 4.5$ s	M1 A1	3.1b 1.1b	Equates an expression for boy's velocity to 5.4 Mark final answer
				[2]		

8	**(c)**		Max distance when $t = 4.5$	**M1**	**3.1b**	recognises that max distance occurs when speeds are equal. Allow for their $t < 5$ from (b) used. Allow for $u = 0$, $v = 5.4$ and $a = 1.2$ in $v^2 = u^2 + 2as$
			boy travels $\frac{1}{2} \times 1.2 \times 4.5^2 = 12.15$ bus travels $4.5 \times 5.4 = 24.3$ m	**M1**	**3.1b**	attempt to use *suvat* or area under the graph and their time $t \leq 5$ to find distance travelled by boy.
			Max distance is 24. 3 – 12.15 =12.15 m	**A1**	**1.1b**	Cao
			Alternative method 1			
			Max distance gained by the bus ahead of the boy is represented by the area of the triangle on velocity-time graph	**M1**		recognises that max distance occurs when the speeds are equal. Allow for their time from (b) used
			Distance $= \frac{1}{2} \times 5.4 \times 4.5 = 12.15$ m	**M1** **A1**		Attempt to find area of the triangle Cao
			Alternative method 2			
			Distance S at time t between $[t < 5]$ $[S =] 5.4t - \frac{1}{2} \times 1.2t^2$	**M1**		Combines expressions from *suvat* equations to find expression for the distance between
			Max occurs when $\frac{dS}{dt} = 5.4 - 1.2t = 0$	**M1**		Equates the derivative of their expression to zero leading to a value for t
			When $t = 4.5$, max distance is 12.15 m	**A1**		Cao
				[3]		

8	**(d)**		Let the time from start to catch the bus be T s			
			Boy's distance is area of trapezium $\frac{1}{2} \times 6(T + T - 5)[= 6T - 15]$	**B1**	**3.1b**	Expression for the total distance for the boy (area method) oe, e.g. sum of two distances $\frac{1}{2} \times 5 \times 6 + (T - 5) \times 6$
			Bus's distance $5.4T$ equate distances $5.4T = 6T - 15$	**M1**	**3.1b**	equates their expression for distance to the distance travelled by the bus and attempt to solve for T
			distance travelled in 25 s is 135 m	**A1**	**1.1b**	Cao. The value for T need not be seen explicitly
			Alternative method (relative speed)			
			At $t = 5$ the boy has travelled $\frac{1}{2} \times 5 \times 6 = 15$ m and the bus $5.4 \times 5 = 27$ m Boy needs to catch up 12 m	**B1**		12 m seen if clear that it is a distance between the boy and the bus
			Boy catches up 12 m at 0.6 ms^{-1} So time is $\frac{12}{0.6}$ [=20 s]	**M1**		Uses relative speed to find the time to catch up
			Total time 25 s gives distance 135 m	**A1**		Cao. 25 s need not be seen explicitly
			Alternative (numerical) method			
			Finds at least one correct distance for boy for $t > 5$	**B1**		
			Working towards the time and distance at which the distances are equal	**M1**		May be awarded for incorrect time and distances eg $0.6t^2$ used for the boy
			For 25 s distance travelled 135 m	**A1**		25 s must be seen as well as 135 m
			No method seen			SC2 for 135 m www where $t = 25$ not seen
				[3]		